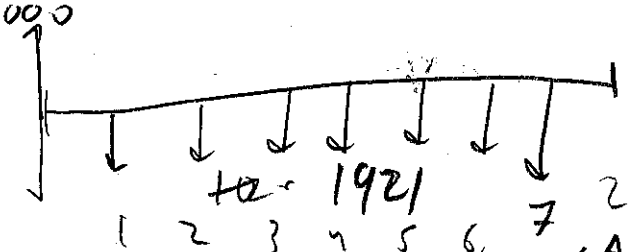
Version 1

Chapter 3

1) You are about to borrow $10,000 from a bank at an interest rate of 8% compounded annually. You are required to make 7 equal annual repayments in the amount of $1921 per year, with the first repayment occurring at the end of year 1. Show the interest payment for year 5, I_5.

10000



$A = F\left[\frac{(1+i)^n - 1}{(1+i)^n}\right]$

1767.32

a) 394.05
b) 396.05
c) 398.05
d) 400.05
e) 402.05

10 000 (F

$A = F(\frac{A}{F}, i, n)$

$1921 = 10000\left(\frac{0.08}{(1+0.08)}\right)$

153.02

2) What will be the amount accumulated in 10 years by the present investment of $15,000 at 9% compounded annually?

a) 35,510.46
b) 35,512.46
c) 35,514.46
d) 35,516.46
e) 35,518.46

$F = P(1+i)^N$

$= 15000(1+0.09)^{10}$

$= 35510.4552$

C

3) For an interest rate of 7% compounded annually, find how much can be lent now if $180,000 will be repaid at the end of 37 years?

a) 14,719.59
b) 14,721.59
c) 14,723.59
d) 14,725.59
e) 14,727.59

$P = F(1+i)^{-N}$

$180000(1+0.07)^{37}$

$= 14725.59$

4) At what rate of interest compounded annually will an investment triple itself in 7 years?

a) 15.5%
b) 16%
c) 16.5%
d) 17%
e) 17.5%

$3 = 1(1+i)^7$

$\ln 3 = 7\ln(1+i)$

$0.15694 = \ln(1+i)$

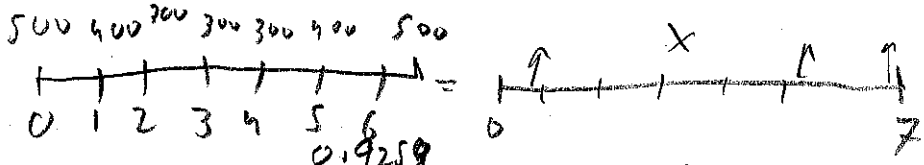
$1.1694 = 1+i$

$i = 0.1699 \approx 17\%$

5) The two cash flow transactions described below are said to be equivalent at 8% compounded annually. Find the unknown value of X that satisfies the equivalence.

Cash flow 1: (t=0) 500, 400, 300, 300, 300, 400, (t=6) 500
Cash flow 2: $X in each period from t=1 to t=7

X$ is between:



a) $413-$416
b) $416-$419
c) $419-$422
d) $422-$425
e) $425-$428

$500 + 400\left(\frac{P}{F}, 8\%, 1\right) + 300\left(\frac{P}{A}, 8\%, 3\right) + 300\left(\frac{P}{F}, 8\%, 2\right) + 300\left(\frac{P}{F}, 8\%, 3\right) + 300\left(\frac{P}{F}, 8\%, 4\right) + 400\left(\frac{P}{F}, 8\%, 5\right) + 500\left(\frac{P}{F}, 8\%, 6\right) = X\left(\frac{P}{A}, 8\%, 7\right)$

0.9259 0.8573 0.7938 0.7350 0.6806 0.6302 5.2064

$2173.53 = X\,5.2064$

$X = 417.4727$

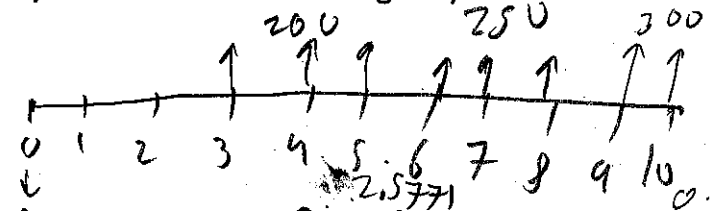
$200(\frac{P}{F}, 8\%, 3) + 200(\frac{P}{F}, 8\%, 4) + 200(\frac{P}{F}, 8\%, 5)$

$+ 250(\frac{P}{F}, 8\%, 6) + 250(\frac{P}{F}, 8\%, 7) + 250(\frac{P}{F}, 8\%, 8)$

$+ 300(\frac{P}{F}, 8\%, 9) + 300(\frac{P}{F}, 8\%, 10) =$

6) Find the present worth of the following cash receipts where i=8% compounded annually (my answer is derived using only 4 interest factors).

$0 from t=0 to t=2
$200 from t=3 to t=5
$250 from t=6 to t=8
$300 from t=9 to t=10



$P = 200(\frac{P}{A}, 8\%, 3)(\frac{P}{F}, 8\%, 2)$

$+ 250(\frac{P}{A}, 8\%, 3)(\frac{P}{F}, 8\%, 5)$

$+ 300(\frac{P}{A}, 8\%, 2)(\frac{P}{F}, 8\%, 8)$

$=$

a) 1002.64
b) 1004.64
c) 1006.64
d) 1008.64
e) 1010.64

7) Ten annual deposits in the amounts of $3000 in t=1, $2800 in t=2, $2600 in t=3, $2400 in t=4, etc. until $1200 in t=10 are made into a fund that pays interest at a rate of 2% compounded annually. Determine the amount in the fund immediately after the 10th deposit.

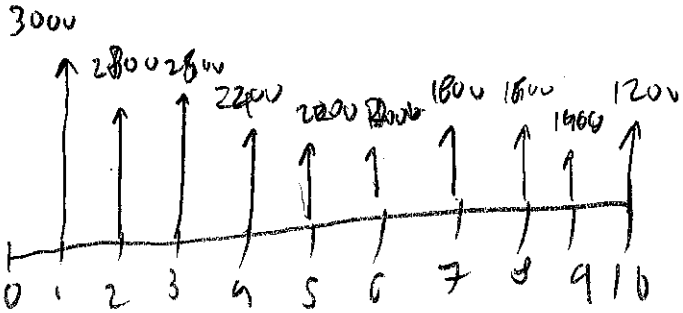
a) 23,351.89
b) 23,353.89
c) 23,355.89
d) 23,357.89
e) 23,359.89

$F = ?$

$[3000 - 200(\frac{P}{A}, 2, 10)](\frac{F}{P}, 2, 10)$

$(\frac{P}{A}, 8\%, 10)$ 8.9826 $\quad$ 1.2190

$= 23352.1114$



23352.11482

0.22482

$20000(F/P, 0.05, 12)$

21233.556 @12

23926.502 @24

$1721.329\left[\frac{0.01(1+0.01)^{12}}{0.01(1+0.01)^{12}-1}\right]$

$= -14.617$

Chapter 4

8) A loan of $20,000 is to be financed over a period of 24 months. The agency quotes a nominal rate of 6% for the first 12 months and a nominal rate of 12% for any remaining unpaid balance after 12 months, both compounded monthly. Based on these rates, what equal end-of-the-month payment for 24 months would be required to repay the loan with interest?

$I = \frac{r}{m} = 0.005$

$P = 20,000$ $N = 24$

$r = 6\%$

$M = 12$

$r_2 = 12\%$

$A = 20000(A/P, 0.5\%, 12)$

$= 1721.329$

a) 921.62

b) 923.62

c) 925.62

d) 927.62

e) 929.62

$I_2 = 0.01$ $A = 1721.329(A/P, 0.01, 12)$

$-20000(A/P, \frac{6\%}{12}, 12) + (A/P, \frac{12\%}{12}, 24)$

9) Mr. Smith wants to buy a new car that will cost $20,000. He will make a down payment in the amount of $5000. He would like to borrow the remainder from a bank at an interest rate of 9% compounded monthly. He agrees to pay off the loan monthly for a period of 3 years.
Immediately after making the 12th payment, Mr. Smith wants to figure out the balance remaining. What is that balance?

Compound = M = 12

$I = \frac{r}{m}$

36

a) B_{12}= $15000(A/P, 9%/12, 36)(F/A, 9%/12, 24)

~~b) B_{12}= $15000(A/P, 9%, 3)(P/A, 0.75%, 12)~~

c) B_{12}= $15000(A/P, 0.75%, 36)(P/A, 0.75%, 24)

d) B_{12}= $15000(A/P, 9%/12, 36)*2/3

e) ~~B_{12}= $15000(A/P, 0.75%, 36)(P/A, 9%, 2)~~

12

10) You borrow $20,000 for buying a new car from a bank at an interest rate of 10% compounded quarterly. This loan will be repaid in 48 equal monthly installments over 4 years. Immediately after the 20th payment, you desire to pay the remainder of the loan in a single payment. How much would this single payment be? (Make sure you round each factor used to only 4 decimals! Your lease payment should be in whole dollar values!)

$M=4 \quad K=52 \quad N=48$

$I=\left(1+\frac{r}{M}\right)^{c}-1$

$\left(1+\frac{0.1}{4}\right)^{\frac{4}{52}}-1 =$

$C=r$

$m=cxk$

$= 1.9\times10^{-3}$

a) 12,595 F_2 $20\,000\left(\frac{F}{P}, 1.9\times10^{-3}, 20\right)$

b) 12,605

c) 12,615 $= 20000\left(1+(1.9\times10^{-3})\right)^{20}$

d) 12,625 $= 20773.87566$

e) 12,635 ≈ 20774

21907.865

1133.98934

≈ 16353.25

11) College Financial Sources, which makes small loans to college students, offers to lend $400. The borrower is required to pay $50 at the end of each week for 3 months. What is the effective weekly interest rate (assume weekly compounding).

$M=52 \quad K=52$

$I=\frac{r}{M}$

a) 7.21%

b) 7.43%

c) 7.56%

d) 7.79%

e) 7.92%

~~Get rid the [illegible]~~

12) What will be the amount accumulated by the present investment of $8534 in 11 years at 8% compounded quarterly.

$M=4$ $K=3$ $C=$

$I=\left(1+\frac{0.08}{4}\right)^{\frac{4}{3}}-1$

a) 20, 390.71
b) 20, 392.71
c) 20, 394.71
d) 20, 396.71
e) 20, 398.71

$F=8534(1+0.15968)^{44}$ $=0.015968$

13) Tony Wu is considering a used automobile. The price, including the title and taxes, is $13,555. Tony is able to make a $3,555 down payment. The balance, $10,000, will be borrowed from his credit union at an interest rate of 9% compounded daily. The loan should be paid in 36 equal weekly payments. Compute the weekly payment.

$M=365$ $C=0.142465$ $I=\left(1+\frac{0.09}{365}\right)^{\frac{52}{365}}-1$

$K=52$

$P=\left(\frac{P}{A},\ ,36\right)$ $=3.51\times10^{-5}$

$A=10000\left(\frac{i(1+i)^{36}}{(1+i)^{36}-1}\right)$

a) $287
b) $290
c) $293
d) $296
e) $299

14) What is the total (dollar) amount of interest Tony has to pay over the life of the loan?

a) 329
b) 332
c) 335
d) 338
e) 341

if you see the guy next to you
circle any answer ~~to~~
get em for me.

Chapter 5

15) A large food processing corporation is considering using laser technology to speed up and eliminate waste in the potato-peeling process. To implement the system, the company anticipates needing $3 million to purchase the industrial strength lasers. The system will **save** $1,550,000 per year in labour and materials. However, it will require an additional annual operating and maintenance cost of $350,000. Annual income taxes will also increase by $150,000. The system is expected to have a 12 year service life and will have a salvage value of $200,000. Find the NPW if the MARR is 8%.

a) 4,992,328
b) 4,992,338
c) 4,992,348
d) 4,992,358
e) 4,238,717

0.3971

$-3000000 + 200000 (F/A, 8, 12)$

$(1550000 - 350000 + 150000)$

$0.98\ 483330$

$(P/A, 8\%, 12)$

5.7466

$= 4957910$

16) The cash flows for a certain project are as follows:

n=0: -1000 (1st cycle)
n=1:+900
n=2:+800
n=3:+500
n=4: -1000, +500 (1st cycle ends, 2nd cycle begins)
n=5: +900
n=6: +800, etc., repeating until 3 cycles are complete.
Find the equivalent annual worth for this project at i=7%.

a) 387.53
b) 392.46
c) 394.81
d) 397.17
e) 399.82

17) A newly constructed bridge costs $5 million. The same bridge is estimated to need renovation every 15 years at a cost of $1 million. Annual repairs and maintenance are estimated to be $200,000 per year. If the interest rate is 7%, determine the capitalized cost of the bridge (it's NPW). You should treat the bridge as having an ***indefinite*** lifespan.

a) 8,421,000
b) 8,425,714
c) 8,960,000
d) 8,981,325
e) 9,224,470

18) A company is currently paying its employees 60 cents per kilometer to drive their own cars on company business. The company is considering supplying employees with cars, which would involve purchasing at $20,000, with an estimated 3 years life, a net salvage value of $5000, taxes and insurance at $1000 per year, and operating and maintenance at $0.20 per kilometer. If the interest rate is 8% and the company anticipates an employee's annual travel to be 24,000 kilometers, then what would be the company's savings per kilometer under the alternative plan?

4800

a) 17 cents
b) 14 cents
c) 10 cents
d) 4 cents
e) 1 cent

0.7938

$-20\,000 + 5000(\frac{P}{F}, 8, 3) + (-1000 - 4800)(\frac{P}{A}, 8, 3)$

2.5771

$= -30978.19$

/440000

21 c

19) Consider an investment project with the following cash flows:

n=0: -6000

n=1: 0

n=2: 4840

n=3: 2662

Compute the IRR for this investment.

a) 8.5%

b) 9%

c) 9.5%

d) 10%

e) 10.5%

20) You are considering a luxury apartment building project that requires an investment of 12,500,000. The building has 50 units. You expect the maintenance cost for the apartment building to be $250,000 the first year, and $300,000 the second year. The maintenance costs will continue to increase by $50,000 in subsequent years. The cost to hire a manager for the building is estimated to be $100,000 per year. After 5 years of operation, the apartment building can be sold for $14,000,000. Assume 90% occupancy during all five years of operation. What is the ***annual*** rent per apartment unit that will provide a return on investment of 8%?

a) 26,368

b) 26,468

c) 26,568

d) 26,668

e) 26,768

21) **<u>Bonus (Professor's choice)</u>**

Consider the following investment project:

n=0: -100

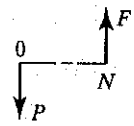
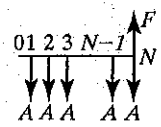
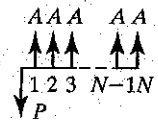
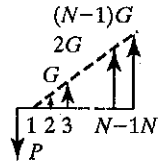
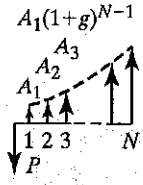
n=1: 256

n=2: -154

Compute the IRR given that MARR=10% (PB approach).

a) 12%
b) 13%
c) 14%
d) 15%
e) 16%

TABLE 3.4 Summary of Discrete Compounding Formulas With Discrete Payments

Type	Factor Notation	Formula	Excel Command	Cash Flow Diagram
SINGLE	Compound amount ($F/P, i, N$)	$F = P(1+i)^N$	$=\text{FV}(i, N, 0, P)$	
	Present worth ($P/F, i, N$)	$P = F(1+i)^{-N}$	$=\text{PV}(i, N, 0, F)$	
EQUAL PAYMENT SERIES	Compound amount ($F/A, i, N$)	$F = A\left[\dfrac{(1+i)^N - 1}{i}\right]$	$= \text{FV}(i, N, A)$	
	Sinking fund ($A/F, i, N$)	$A = F\left[\dfrac{i}{(1+i)^N - 1}\right]$	$=\text{PMT}(i, N, 0, F)$	
	Present worth ($P/A, i, N$)	$P = A\left[\dfrac{(1+i)^N - 1}{i(1+i)^N}\right]$	$=\text{PV}(i, N, A)$	
	Capital recovery ($A/P, i, N$)	$A = P\left[\dfrac{i(1+i)^N}{(1+i)^N - 1}\right]$	$=\text{PMT}(i, N, P)$	
GRADIENT SERIES	Linear gradient Present worth ($P/G, i, N$)	$P = G\left[\dfrac{(1+i)^N - iN - 1}{i^2(1+i)^N}\right]$		
	Annual worth ($A/G, i, N$)	$A = G\left[\dfrac{(1+i)^N - iN - 1}{i\left[(1+i)^N - 1\right]}\right]$		
	Geometric gradient Present worth ($P/A_1, g, i, N$)	$P = \begin{cases} A_1\left[\dfrac{1-(1+g)^N(1+i)^{-N}}{i-g}\right] \\ A_1\left(\dfrac{N}{1+i}\right), (\text{if } i = g) \end{cases}$